

Heathrow Helper

Tech stack reference — languages, libraries, APIs and infrastructure used to build heathrow-helper.vercel.app.

Languages

Python 3.12	Backend logic — <code>bot.py</code> (~2,000 lines), <code>app.py</code> (Flask routes)
HTML (Jinja2)	Page templates — index, about, privacy, terms, contact
CSS3	Styling — tokens, base, chat, pages, dark
JavaScript (vanilla)	Chat UI — <code>static/js/chat.js</code> , no framework
JSON	Static reference data — 18 files in <code>/data</code>

Backend — Python libraries (`requirements.txt`)

Flask 3.0+	Web framework, routing, templates
gunicorn	Production WSGI server (used by Render)
requests	HTTP client for Heathrow + AviationStack APIs
python-dotenv	Loads <code>.env</code> for the AviationStack key locally
langdetect	Detects the user's input language
deep-translator	Translates non-English queries to English

Frontend — runtime libraries (CDN)

marked v12	Markdown → HTML rendering in chat replies
DOMPurify v3	XSS sanitisation of bot output
Geist & Geist Mono	Typography (Google Fonts)

Data sources (live APIs)

Heathrow Data Platform <code>api-dp-prod.dp.heathrow.com</code>	Live departures/arrivals, status, codeshares, security & immigration waits
AviationStack	Fallback flight lookup for non-LHR flights

Deployment & infrastructure

Vercel	Primary host — <code>heathrow-helper.vercel.app</code> (Python serverless)
Render	Backup host — <code>heathrow-helper.onrender.com</code>
Vercel CLI	Deploy via <code>vercel deploy --prod</code>
Git + GitHub	Source control — <code>rec0334/Heathrow-Helper</code>
npm	Used only to install the Vercel CLI globally

Configuration files

vercel.json	Vercel — rewrites all routes to api/index.py
Procfile	Render — web: gunicorn app:app
runtime.txt	Render — pins python-3.12.4
.vercelignore	Vercel — excludes .env, render configs, tests
requirements.txt	Both hosts — Python dependency list

Local dev tools

VS Code / Claude Code	Editing
Playwright	Mobile + dark-mode screenshot testing
curl	API smoke-testing live endpoints
test_live.py	Live end-to-end tests

One-line summary

Python (Flask) backend serving a vanilla-JS chat UI, deployed on Vercel as a Python serverless function, with live Heathrow API + AviationStack data and ~18 JSON files of static reference content.

No build step. No framework. No database — by design, to keep cold start under 1 second and the codebase scannable in one read.